

CONTROL SYSTEM
DIGITAL ASSIGNMENT 3
SLOT : C1 + TC1

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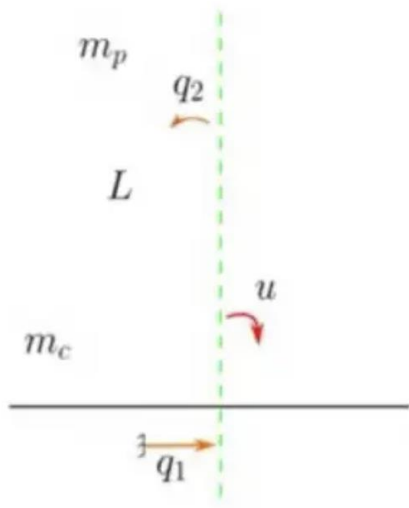
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CONTROL OF INVERTED PENDULUM SYSTEM

An inverted pendulum system is a classic control engineering problem that involves balancing a pendulum in an upright position using a controlled force. The system is naturally unstable and requires continuous feedback control to maintain equilibrium.

Control Objective:

- The goal is to keep the pendulum balanced upright.
- This is a complex control problem because any movement of the cart affects the pendulum and vice versa.



A simplified model diagram

It represents the inverted pendulum system with a cart of mass m_c and a pendulum of mass m_p

- The pendulum is attached to the cart via a massless arm of length L .
- The key variables:
 - q_1 : The cart's displacement (horizontal movement).
 - q_2 : The pendulum's angle (rotation from the vertical axis).
- Control input u is a force applied to the cart to maintain balance and move the system.
- The goal is to control the cart's motion to keep the pendulum balanced upright using feedback from q_1 and q_2 .

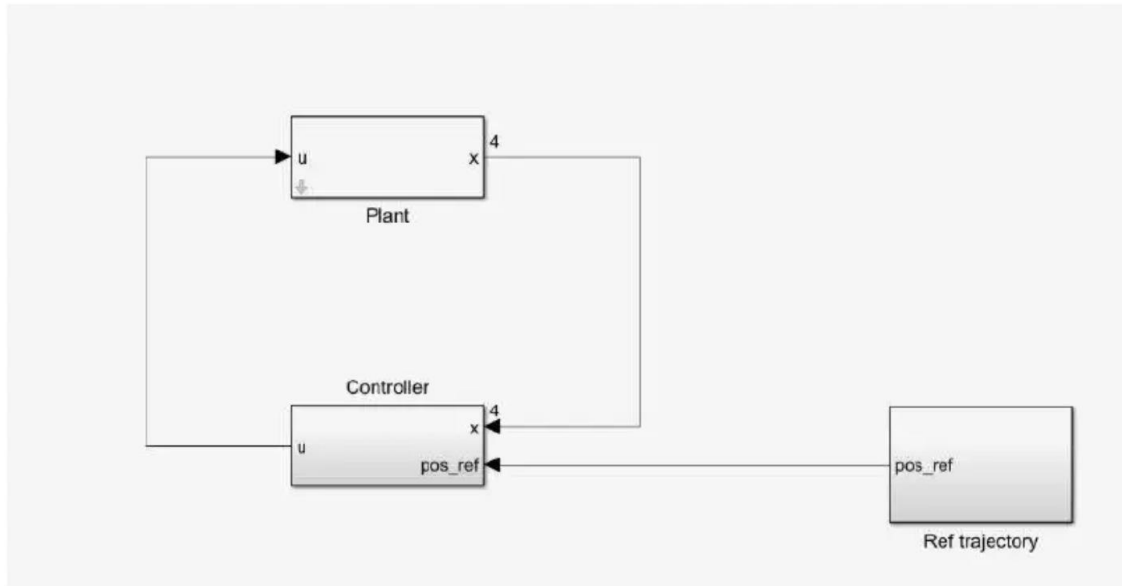
Applications of using inverted pendulum system



Fig 1. Self-balanced Robot with the use of inverted pendulum control system

The robot maintains balance by continuously adjusting its wheels based on sensor feedback. An IMU detects deviations from the upright position, and a PID controller processes this data to generate corrective motor movements. The proportional term corrects errors, the integral term eliminates steady-state error, and the derivative term predicts future errors. These adjustments ensure real-time stability, allowing the robot to remain upright. This control mechanism mimics an inverted pendulum system, making the robot self-sustaining and adaptable to disturbances while maintaining balance efficiently.

System Design



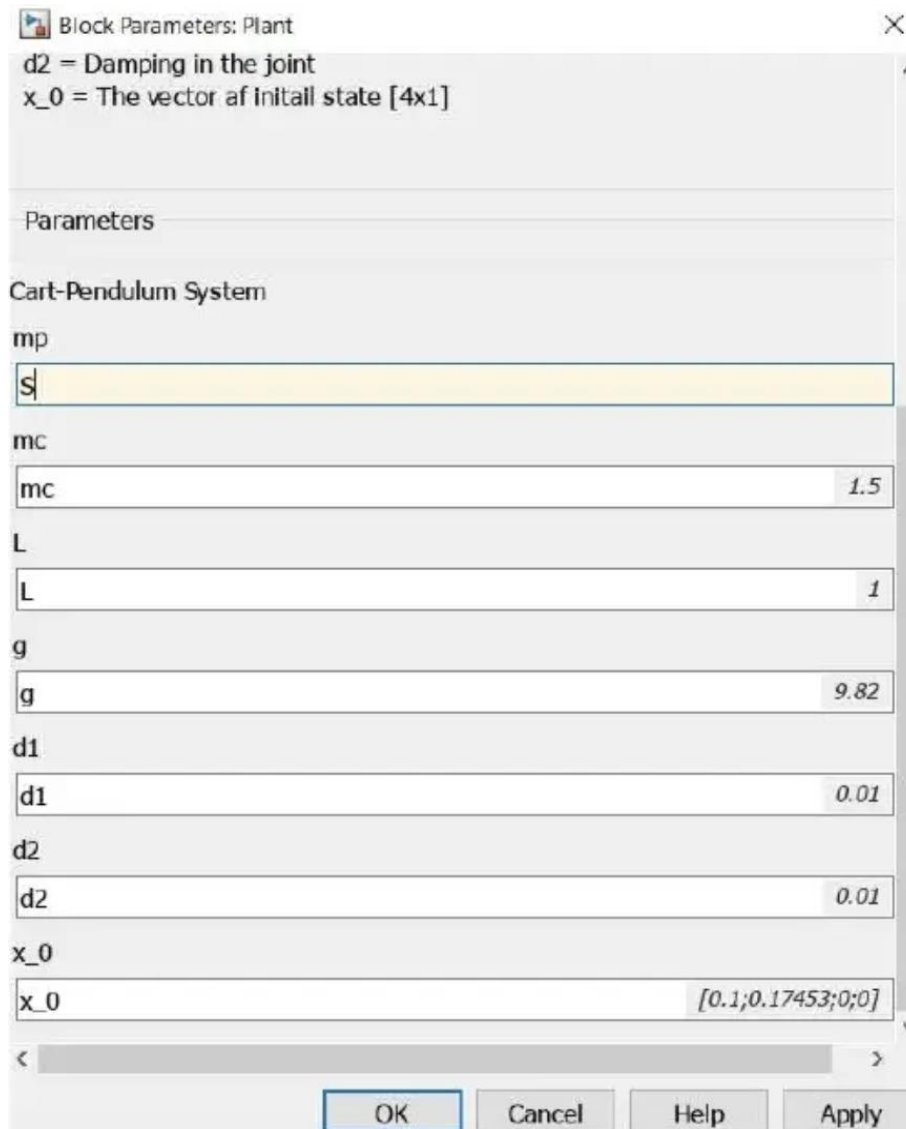
This project involves the simulation and control of a cart-pendulum system using MATLAB Simulink. The system consists of a cart with a pendulum attached to it, forming a nonlinear dynamic model. The primary goal is to develop a control system that stabilizes the pendulum while tracking a reference trajectory.

System Model (Plant block):

The plant model used in the simulation represents the nonlinear dynamics of the cart-pendulum system. The block parameters are as follows:

- m_c (Mass of the cart): 1.5 kg
- m_p (Mass of the pendulum): Variable (to be determined)
- L (Length of the pendulum): 1 m
- g (Gravity acceleration): 9.82 m/s^2

- **d1** (Damping of the cart displacement): 0.01
- **d2** (Damping in the joint): 0.01
- **x_0** (Initial state vector): [0.1; 0.17453; 0; 0]



Block Parameters: Plant

d2 = Damping in the joint
x_0 = The vector of initial state [4x1]

Parameters

Cart-Pendulum System

mp

S

mc

mc 1.5

L

L 1

g

g 9.82

d1

d1 0.01

d2

d2 0.01

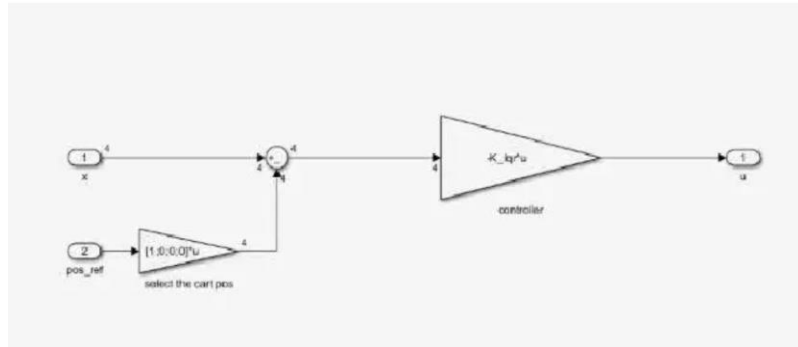
x_0

x_0 [0.1;0.17453;0;0]

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Controller Block:



1. Inputs to the Controller

- **x(State Vector Input, 4x1):** This represents the current state of the system, which includes cart position, cart velocity, pendulum angle, and pendulum angular velocity.
- **posref (Reference Position):** This provides the desired position for the cart.

2. Reference Position Processing

- The posref input is multiplied by $[1; 0; 0; 0]$, which selects only the cart position component from the state reference. This means the controller will primarily focus on keeping the cart at the desired reference position while stabilizing the pendulum.

3. Error Calculation

- The reference cart position is subtracted from the actual cart position. This generates the error signal, which represents the difference between the desired and actual cart positions.

4. Control Law: State Feedback Controller

- The error signal is fed into a **state feedback controller** of the form:

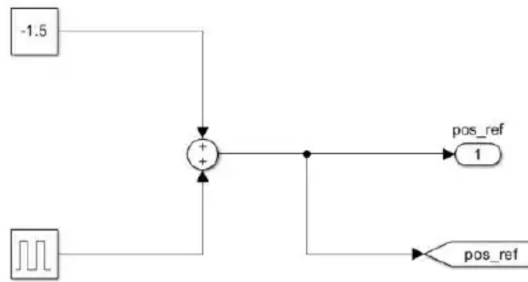
$$u = -K \cdot x$$

where K is a precomputed state feedback gain matrix (denoted as K_LQR in the block). This gain is usually designed using Linear Quadratic Regulator (LQR) techniques to ensure optimal control performance.

5. Output: Control Input u

- The computed control signal u is then sent to the plant (cart-pendulum system) to adjust the forces acting on the cart, thereby achieving the control objectives.

Reference Trajectory:



The reference trajectory subsystem defines the desired movement of the cart in a cart-pendulum system. This trajectory serves as the input for the controller, guiding the cart to follow a pre-defined motion.

This provides input between one and half to minus one and half.

1. A Constant Block (-1.5)

- This provides a constant bias to the reference trajectory.
- It shifts the trajectory downward by 1.5 units.

2. A Signal Generator Block

- This generates a periodic waveform, possibly a square or sine wave.
- The waveform defines how the cart moves over time (e.g., oscillating back and forth).

3. A Summation Block

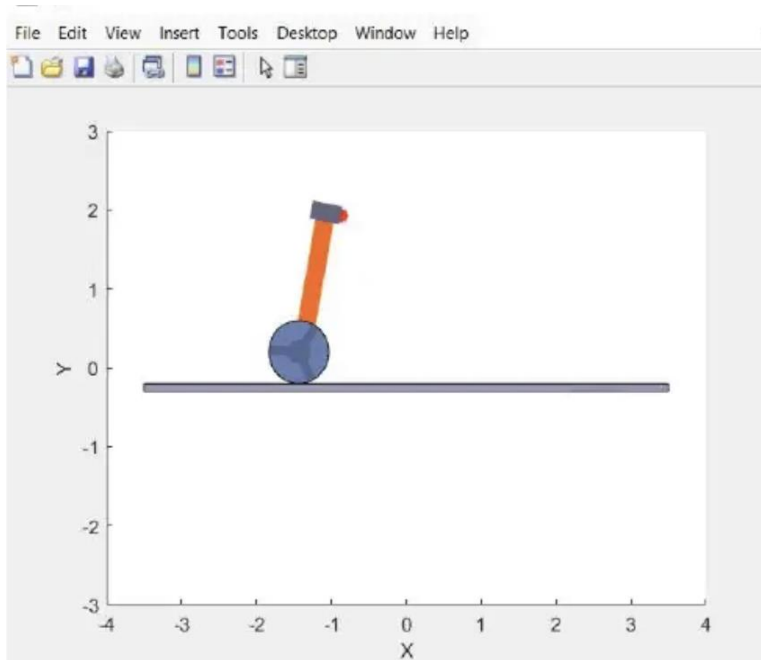
- The sum of the constant value (-1.5) and the generated waveform gives the final reference trajectory.
- This determines the cart's desired motion.

4. Output Signal (pos_ref)

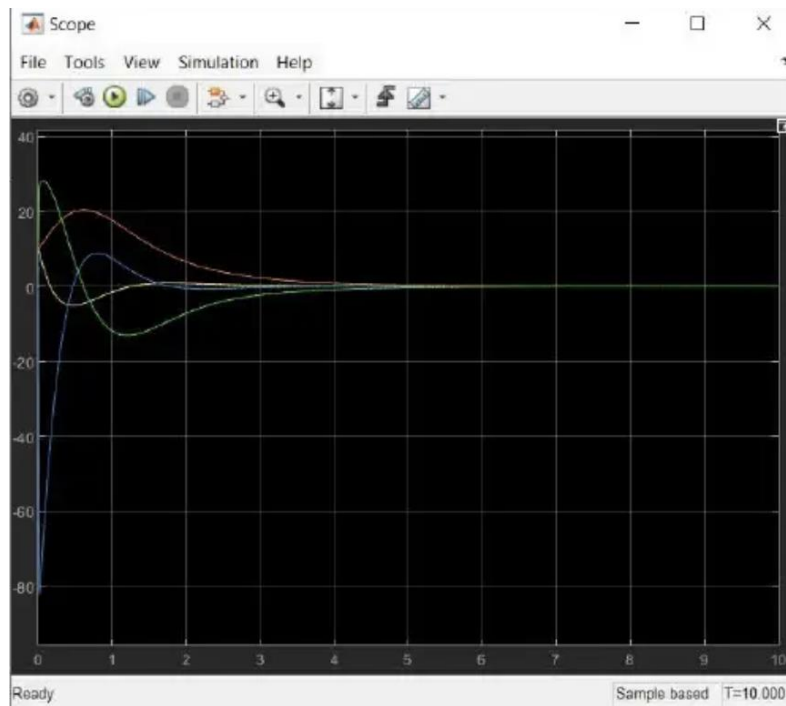
- This serves as the desired cart position.
- The controller will attempt to minimize the error between the actual cart position and this reference.

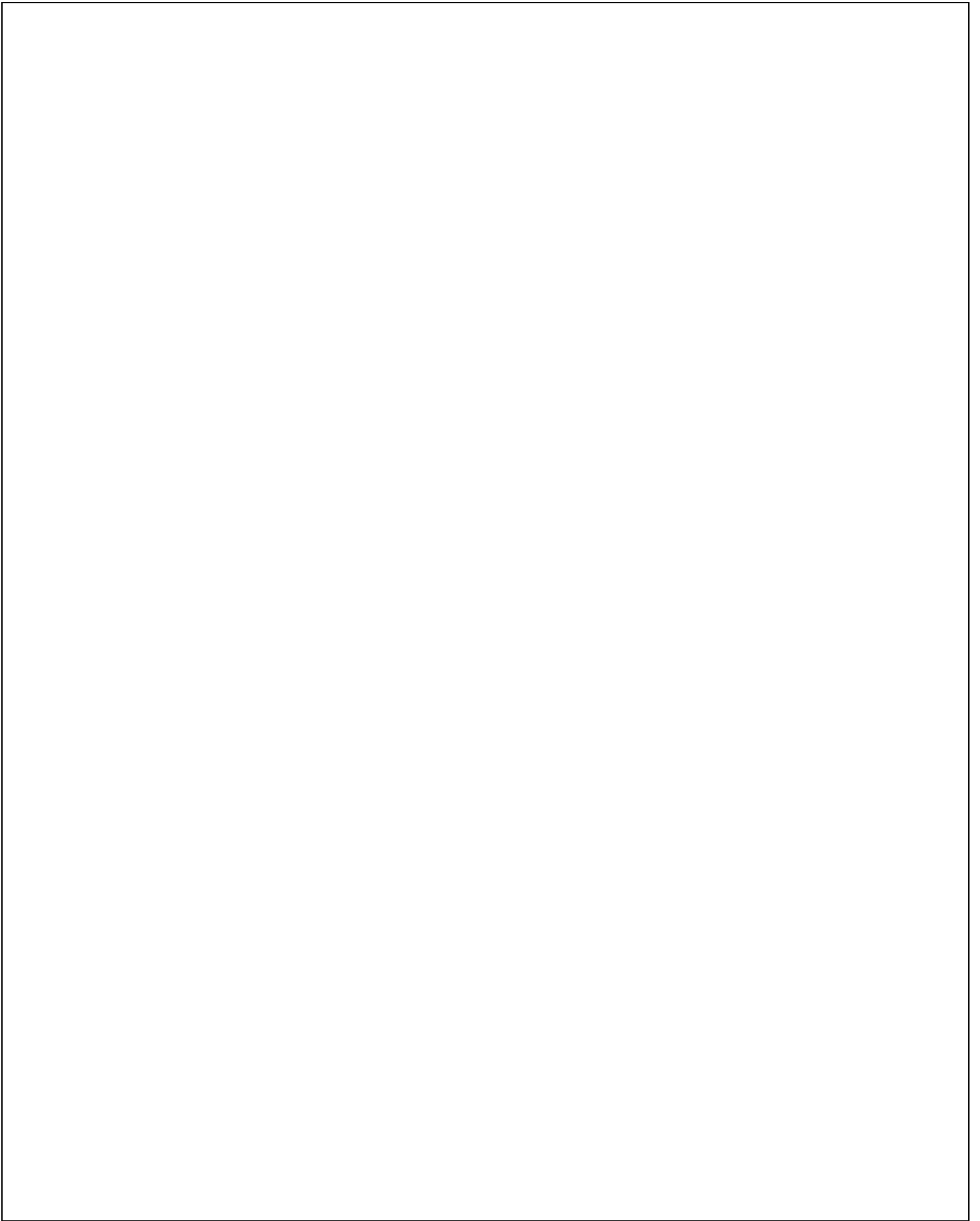
Implementation and Result:

Model:



Graph of the closed loop inverted pendulum system:

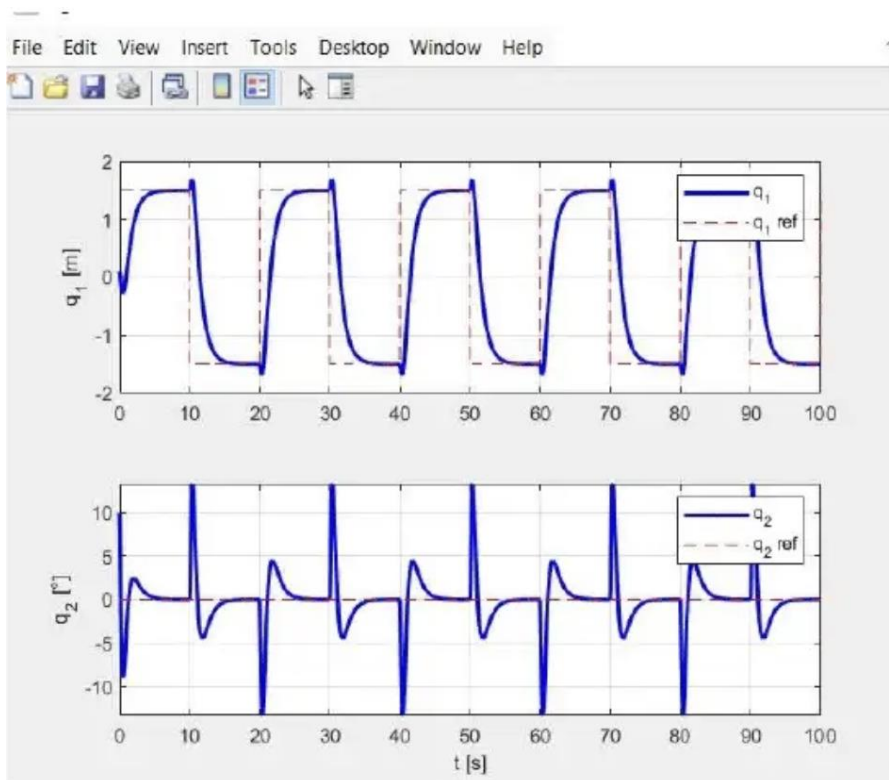




Graph from reference trajectory:

The graph represents the position and angle of the cart.

- The top plot shows q_1 which is the cart position in meters.
- The bottom plot shows q_2 which is the pendulum angle in degrees.



Conclusion

The Simulink modelling of the inverted pendulum system effectively demonstrates control and stability principles. Key takeaways include:

- ***System Instability***: The inverted pendulum is inherently unstable and requires continuous control.
- ***PID Controller***: A PID-based control strategy stabilizes the system by adjusting the cart's motion based on feedback.
- ***Simulink Advantages***: The graphical, block-based environment allows for easy modelling, simulation, and parameter tuning.
- ***Control Performance***: Proper PID tuning ensures a fast, stable response while preventing excessive oscillations.
- ***Real-World Applications***: This model is applicable in self-balancing robots, aerospace, and robotics.
- ***Future Enhancements***: Alternative control strategies, such as LQR (Linear Quadratic Regulator) or Fuzzy Logic Control, can improve performance.